

PageRank I

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Reputation Systems

Reputation System

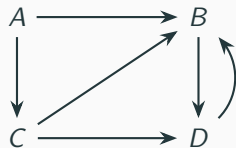
- Exists in many social networks from hotel reviews to search.
- Example 1: TripAdvisor allows users to rate hotels and restaurants and search for ones in a particular location.
- Example 2: Google uses a reputation system to rank web pages.

Naive PageRank

- Even the most obscure search can return a lot of results.
- Search engine must order the results in a way that is useful to the user.
- Example: “Used cars in New York City”.
- Page Rank is a method for ranking web pages.

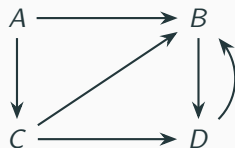
Naive PageRank

- If A links to B , then A is said to think highly of B .
- If many pages link to B , then B 's reputation is high.



Naive PageRank

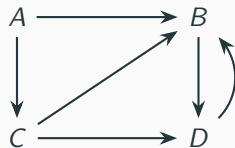
- Let d_v be the number of outgoing links from page v .
- Let π_v be the pagerank of page v , to be defined.
- Example: $d_C = 2$.



Naive PageRank

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$$\pi_v = \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u}$$



Definition

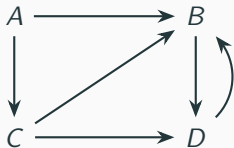
$$\pi_v = \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u}$$

- Constraint: $\sum_{v \in V} \pi_v = 1$ and $\pi_v \geq 0$ for all $v \in V$.
- Intuition: if u has an outgoing edge then u contributes $1/d_u$ to the pagerank of v .

Random Walk Interpretation

Random walk interpretation

- One can think of this as a monkey surfing the web.
- At each step, the monkey follows a link from the current page to the next page.



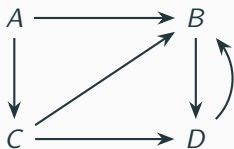
π_B can be interpreted as the fraction of time the monkey spends at B .

$$\pi_B = \frac{1}{2}\pi_A + \frac{1}{2}\pi_C + \pi_D.$$

Computing PageRank

PageRank's naive computation

Solve a system of linear equations.



$$\pi_A = 0$$

$$\pi_B = \frac{1}{2}\pi_A + \frac{1}{2}\pi_C + \pi_D$$

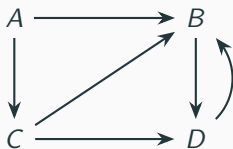
$$\pi_C = \frac{1}{2}\pi_A$$

$$\pi_D = \pi_B + \frac{1}{2}\pi_C$$

$$\pi_A + \pi_B + \pi_C + \pi_D = 1$$

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This gives $\pi_A = 0$, $\pi_B = \frac{1}{2}$, $\pi_C = 0$, $\pi_D = \frac{1}{2}$.

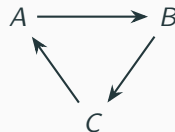
PageRank's naive computation

- The number of equations is the number of nodes n .
- $O(n^3)$ time using Gaussian elimination or matrix inversion.
- Infeasible for billions of web pages.
- We will deal with this later.

Flaws with Naive PageRank

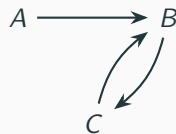
Flaws with naive PageRank (PageRank steal)

- A webpage cannot force other webpages to link to it.
- However, it can change its pagerank by changing its outgoing links.
- In the above, the pagerank of each page is $1/3$.
- Let see what happens if C changes its link to A to B .



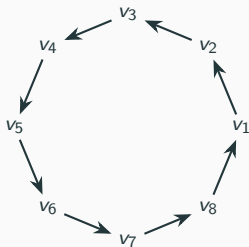
Flaws with naive PageRank (PageRank steal)

- Since A has no link to it, its pagerank is now 0.
- The pagerank of B and C is now $1/2$.
- Thus, C can “steal” pagerank from A .



Flaws with naive PageRank (PageRank steal)

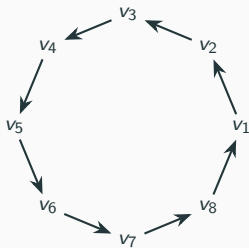
We can magnify the effect of this by creating a longer cycle.



All pageranks are equal which is $1/8$.

Flaws with naive PageRank (PageRank steal)

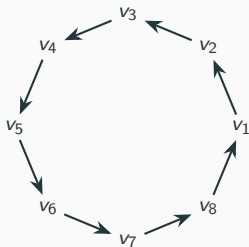
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Suppose v_2 redirects its link to v_3 toward v_1 . What happens?

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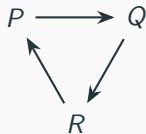
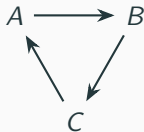
We can magnify the effect of this by creating a longer cycle.



Suppose v_2 redirects its link to v_3 toward v_1 . What happens? The new pageranks of $v_3, v_4, \dots, v_8$ are now 0. The new pagerank of v_1 and v_2 are now $1/2$.

Flaws with naive PageRank (Disconnected Components)

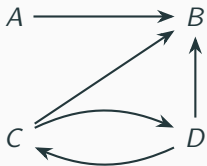
An even more serious flaw is that pagerank may not be well-defined.



- In the above, $\pi_A = \pi_B = \pi_C$ and $\pi_P = \pi_Q = \pi_R$.
- Recall:
$$\pi_A + \pi_B + \pi_C + \pi_P + \pi_Q + \pi_R = 1.$$
- $\pi_A = \pi_B = \pi_C = \pi_P = \pi_Q = \pi_R = 1/6$ is a solution.
- $\pi_A = \pi_B = \pi_C = 0$ and $\pi_P = \pi_Q = \pi_R = 1/3$ is another solution.

Flaws with naive PageRank (Dangling nodes)

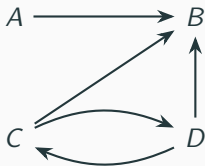
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- $\pi_A = 0$; $\pi_B = \pi_C/2 + \pi_D/2$;
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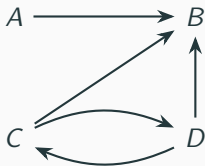
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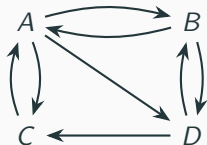
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- This implies that $\pi_C = \pi_D = 0$ and hence $\pi_B = 0$. But this violates the constraint $\pi_A + \pi_B + \pi_C + \pi_D = 1$.
- This means the solution might not even exist.

Matrix Formulation

Transition Matrix



The transition matrix with
 $A = 1, B = 2, C = 3, D = 4$:

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}.$$

$M_{ij} = 1/d_j$ if there is an edge from j to i , and
0 otherwise.

Power Iteration

Suppose the surfer is at a random page at time 0. Then $\pi^{(0)} = [1/4, 1/4, 1/4, 1/4]^T$.

The distribution at time 1 is $\pi^{(1)} = M\pi^{(0)}$:

$$\pi^{(1)} = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 3/8 \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix} \approx \begin{bmatrix} 0.375 \\ 0.208 \\ 0.208 \\ 0.208 \end{bmatrix}.$$

Power Iteration (cont.)

Continuing to apply $\pi^{(t)} = M\pi^{(t-1)}$:

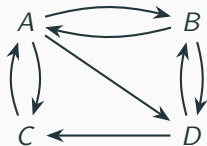
$$\pi^{(2)} = \begin{bmatrix} 15/48 \\ 11/48 \\ 11/48 \\ 11/48 \end{bmatrix} \approx \begin{bmatrix} 0.313 \\ 0.229 \\ 0.229 \\ 0.229 \end{bmatrix}$$

$$\pi^{(3)} = \begin{bmatrix} 11/32 \\ 7/32 \\ 7/32 \\ 7/32 \end{bmatrix} \approx \begin{bmatrix} 0.344 \\ 0.219 \\ 0.219 \\ 0.219 \end{bmatrix}$$

$$\pi^{(4)} = \begin{bmatrix} 63/192 \\ 43/192 \\ 43/192 \\ 43/192 \end{bmatrix} \approx \begin{bmatrix} 0.328 \\ 0.224 \\ 0.224 \\ 0.224 \end{bmatrix}$$

The distribution approaches a stationary π with $\pi = M\pi$. This gives the same definition as before.

Stationary Distribution



$M\pi = \pi$ gives:

$$\pi_A = \frac{1}{2}\pi_B + \pi_C$$

$$\pi_B = \frac{1}{3}\pi_A + \frac{1}{2}\pi_D$$

$$\pi_C = \frac{1}{3}\pi_A + \frac{1}{2}\pi_D$$

$$\pi_D = \frac{1}{3}\pi_A + \frac{1}{2}\pi_B$$

Equivalence of Definitions

The transition matrix encodes the same relation:

$$(M\pi)_i = \sum_j M_{ij} \pi_j = \sum_{j \rightarrow i} \frac{1}{d_j} \pi_j = \sum_{j \rightarrow i} \frac{\pi_j}{d_j} = \pi_i.$$

Thus the PageRank vector π is a stationary distribution ($\pi = M\pi$) of the random walk on the graph.

Conditions for Unique PageRank

The distribution approaches a stationary distribution π such that $\pi = M\pi$.

From Markov process theory, such π exists, is unique, and power iteration converges to it if:

- The graph is strongly connected (irreducible).
- The graph has no dangling nodes (dead ends).
- The graph is aperiodic: GCD of all cycle lengths is 1.

The ϵ -teleportation fix ensures all three hold simultaneously.